

SHANE DUBE

Johannesburg, GP • 071 330 9358 • shanedube11@gmail.com • github.com/filesOfShane

PROFESSIONAL SUMMARY

Data Science graduate with hands-on experience across the full ML lifecycle; from data pipeline construction and feature engineering through model training, evaluation, and deployment. Skilled in Python, scikit-learn, TensorFlow, and SQL. Experienced working in team-based research environments and independently building production-structured pipelines. Passionate about applying data-driven solutions to real-world business and infrastructure challenges.

EDUCATION

BSc Information Technology — Major: Data Science

Eduvos | 2022 – 2025

Completed

PROJECTS & EXPERIENCE

Alzheimer's & Dementia Classification | Group Research Project

Jul 2025 – Dec 2025

- Designed and compared multiple ML models (Random Forest + CNN hybrid) to classify dementia severity from clinical datasets and MRI scans, achieving
- 92% accuracy on structured clinical data and 85% on MRI image data using a two-phase hybrid modelling architecture (RF + CNN).
- Built end-to-end data pipelines: missing value imputation, OneHotEncoding, StandardScaler, and SMOTE for class balancing.
- Engineered feature importance visualisations to surface key cognitive and clinical risk indicators — translating model outputs into actionable diagnostic insights.
- Deployed interactive Streamlit dashboard for medical practitioners, outputting patient risk levels (Low/Medium/High) with model confidence scores.
- Collaborated in a multi-member team following a structured research methodology; project hosted on GitHub with full version control.

Tools: Python, Pandas, scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, Streamlit, Git

Customer Churn Prediction System | Individual Project

Jan 2026 – Present

- Constructed a production-structured ML pipeline (with separate preprocessing, training, and evaluation scripts) to predict churn across 440,000+ customer records.
- Performed EDA to identify key churn drivers — usage frequency, support call volume, payment delays — and built feature encoding pipelines for model-ready data.
- Achieved ROC-AUC of 0.91 (Logistic Regression baseline) and 0.95 (Random Forest); serialised models with joblib for deployment.
- Built a Streamlit front-end, demonstrating full MLOps lifecycle from training to serving.
- Demonstrated business value: predictive retention targeting to reduce churn-related revenue loss.

Tools: Python, Pandas, scikit-learn, Streamlit, Git

TECHNICAL SKILLS

Languages Python (Pandas, NumPy), SQL

ML / AI scikit-learn, TensorFlow, Keras — Random Forest, CNNs, Logistic Regression

MLOps Model evaluation (ROC-AUC, confusion matrix, precision-recall)

Data Engineering Feature engineering, data cleaning, EDA

Visualisation & BI Matplotlib, Seaborn, Power BI, Streamlit dashboards

Tools Git, GitHub, VS Code

CORE COMPETENCIES

Problem-solving mindset • Clear communication of complex ideas • Independent and collaborative work •
Attention to detail • Continuous learning • End-to-end project ownership